

Deepak Mallampati

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Kent, OH - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) building production conversational AI agents and RAG systems in healthcare -- the exact vertical NICE's Agentic AI Transformation team targets. Track record of managing the full agent lifecycle from POC to production: ~25% improved agent stability, >30% hallucination reduction, >90% retrieval-grounded alignment. High-agency builder with independent shipped products (Manga Lens Chrome extension, Agentic Healthcare Claims pipeline). Experienced with agent evaluation frameworks, orchestration tools, RAG pipelines, prompt engineering, and HIPAA-conscious governance. React and TypeScript proficient.

CORE COMPETENCIES

Conversational AI Agent Development • Agent Lifecycle Management (POC to Production) • RAG Pipeline Design • Healthcare AI (HIPAA-conscious) • Prompt Engineering & Evaluation • LangChain & LlamaIndex • Python, TypeScript, React • FastAPI / Docker • Hallucination Guardrails • High-Agency Independent Building

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting | USA

- Built **production conversational AI agents** for multi-step healthcare workflows (eligibility checks, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability ~25% and reduced hallucinations >30% -- directly applicable to NICE's enterprise agent deployment.
- Designed **RAG systems** for clinical knowledge retrieval; improved contextual retrieval precision ~35%, reduced irrelevant retrieval >30%, and achieved >90% retrieval-grounded response alignment.
- Managed the **full agent lifecycle from POC through production hardening**: evaluation pipeline design, grounding threshold calibration, HIPAA de-identification, audit trails, FastAPI packaging with Docker.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient risk scoring; improved recall 15-20% on high-risk categories while holding precision >90%.
- Maintained HIPAA-conscious data governance across healthcare AI systems in a regulated enterprise context.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP, Video Analytics | Hyderabad, India

- Built transformer-based video summarization system; reduced manual review time 60-70% and achieved ~85% alignment with human-curated highlights.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~30% and raised contextual accuracy >85%.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP | Hyderabad, India

- Optimized SQL/T-SQL stored procedures; reduced query latency ~25%.
- Built CI/CD pipelines; reduced deployment errors >30%.
- Designed Grafana performance dashboards; cut incident recurrence ~25%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake Normalization to Fraud Risk Scoring) with schema-validated JSON contracts between agents, RAG-grounded CPT/ICD validation, and explainable risk scoring with audit-ready reasoning traces. Full agent lifecycle owned end-to-end in a regulated healthcare context.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG, multi-agent orchestration

Manga Lens - Chrome Extension for Real-Time AI Translation shipped, Chrome Web Store

Built and shipped independently: Manifest V3, 4 AI vision providers (including Claude Sonnet), multi-section capture pipeline, 7-day cache, 29-site support. Demonstrates high-agency independent product delivery -- "natural builder's mentality" in practice.

TypeScript, Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama

Agentic Pixel Character Synthesis & Animation Engine

LLM-based agent orchestrator decomposing high-level prompts into generation tasks, with Stable Diffusion fine-tuning, LoRA identity consistency, and ControlNet pose conditioning.

PyTorch, HuggingFace Diffusers, FastAPI, Docker, LLM orchestration

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: Python, TypeScript, React, FastAPI, Flask, SQL

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, evaluation pipelines, guardrails, orchestration tools

ML: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, embeddings, vector search

Infra: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Conversational AI Agents, Enterprise AI, Financial Services AI, Regulated Industry Deployment